

NumPy Mastery — 100 Questions

Function Reference, Tasks, and Programs

Part 1 lists the core NumPy functions you should know exist. Part 2 has 60 short tasks. Part 3 has 40 full programs, ending at very advanced. Basic to very advanced, in order.

PART 1 — NumPy Function Checklist

Array Creation

array, zeros, ones, full, eye, identity, arange, linspace, logspace, empty, zeros_like, ones_like, full_like

Array Attributes

shape, ndim, size, dtype, itemsize, nbytes, astype

Indexing & Selection

basic slicing, fancy indexing, boolean indexing, where, take, select, nonzero, argwhere

Reshaping

reshape, flatten, ravel, transpose, swapaxes, moveaxis, expand_dims, squeeze, newaxis

Combining & Splitting

concatenate, vstack, hstack, dstack, column_stack, stack, split, hsplit, vsplit, array_split

Elementwise Math

add, subtract, multiply, divide, power, sqrt, exp, log, abs, round, floor, ceil, clip

Aggregation

sum, mean, median, std, var, min, max, argmin, argmax, cumsum, cumprod, prod

Sorting & Searching

sort, argsort, unique, searchsorted, count_nonzero

Linear Algebra

dot, matmul, linalg.inv, linalg.det, linalg.eig, linalg.svd, linalg.norm, linalg.solve, linalg.matrix_rank, trace

Random

random.seed, random.rand, random.randn, random.randint, random.choice, random.shuffle,
random.permutation, random.normal, random.uniform, random.binomial

Logic & Comparison

all, any, isnan, isinf, isfinite, logical_and, logical_or, logical_not, array_equal

Broadcasting & Misc

tile, repeat, meshgrid, einsum, vectorize, apply_along_axis, bincount, digitize, histogram, corrcoef, cov, pad, roll, diag

PART 2 — Tasks (60)

1. Create a 1D array of numbers from 0 to 20.
2. Create a 2D array of shape 4x5 filled with ones.
3. Create an identity matrix of size 6.
4. Create an array of 15 evenly spaced values between 10 and 50.
5. Print the shape, size, and dtype of a given array.
6. Convert a nested Python list into a 2D NumPy array.
7. Create a 3x3 array filled with the value 9.
8. Reshape a 1D array of 20 elements into a 4x5 matrix.
9. Change the dtype of an array to float64.
10. Create an array of 10 zeros and then set the 5th element to 100.
11. Extract every 3rd element from a 1D array.
12. Reverse a 2D array along its rows.
13. Extract the last row and last column of a matrix.
14. Get all values from an array that are divisible by 3.
15. Replace all values greater than 50 in an array with 50.
16. Extract a 3x3 sub-matrix from the center of a 6x6 matrix.
17. Find the indices of all elements equal to a given value.
18. Select rows of a matrix where the value in the first column is even.
19. Select rows at index positions 1, 4, and 6 from a matrix.
20. Set all values below 0 in an array to 0 and above 100 to 100.
21. Flatten a 3D array of shape (2,3,4) into 1D.
22. Reshape a (60,) array into (3,4,5).
23. Stack two 1D arrays vertically and then horizontally.
24. Concatenate two 2D arrays row-wise and column-wise.
25. Split an array of 20 elements into 4 equal parts.
26. Transpose a 3x5 matrix.
27. Convert a 1D array of shape (10,) into shape (10,1).
28. Swap the first and last axis of a 3D array.
29. Combine four 1D arrays into a single 2D array using columns.
30. Repeat a 1D array 4 times to build a 2D array.

31. Add a 1D array to each row of a 2D array.
32. Multiply each column of a matrix by a different value from a 1D array.
33. Subtract the mean of each column from that column, across a whole matrix.
34. Compute the distance between every point in array A and every point in array B.
35. Write a function that returns $1/(1+e^{-x})$ for an input array.
36. Write a function that converts a batch of raw scores into probabilities that sum to 1 per row.
37. Write a function that zeroes out all negative numbers in an array.
38. Compute the elementwise min and max of two arrays.
39. Convert an integer array of class labels into a one-hot encoded 2D array.
40. Compute the matrix product of a 4x3 and a 3x2 matrix.
41. Compute mean, median, std and variance of every column in a matrix.
42. Find the minimum and maximum value of every column in a matrix.
43. Scale every column of a matrix into the range 0 to 1.
44. Scale every column of a matrix to have mean 0 and std 1.
45. Count the number of NaN values in each column of a matrix.
46. Replace every NaN in a column with that column's average.
47. Find outlier values in a column using the IQR rule.
48. Compute the correlation between every pair of columns in a matrix.
49. Compute the covariance matrix of a dataset.
50. Find the most frequently occurring value in an array.
51. Compute a running (cumulative) sum of an array.
52. Shuffle the rows of a feature matrix and a label array together, keeping them aligned.
53. Split a dataset into 80% training data and 20% testing data.
54. Count how many samples belong to each class in a label array.
55. Convert a column of continuous ages into 4 discrete age-group bins.
56. Compute the inverse and determinant of a 3x3 matrix.
57. Solve a system of 3 linear equations with 3 unknowns.
58. Build Linear Regression weights using the Normal Equation on a small dataset.
59. Compute the eigenvalues and eigenvectors of a 3x3 covariance matrix.
60. Compute the L1 and L2 norm of a vector.

PART 3 — Programs (40)

61. Write a program that generates a random dataset of 100 samples with 3 features, then normalizes every feature to the 0-1 range.

62. Write a program that generates a random dataset with missing values, detects them, and fills every missing value with its column mean.
63. Write a program that splits a random dataset of 200 samples into training and testing sets in an 80/20 ratio.
64. Write a program that generates 500 random 2D points, computes the distance of every point from a given center point, and finds the 5 closest points.
65. Write a program that implements the sigmoid function and prints its output for x ranging from -10 to 10.
66. Write a program that implements a single dense layer forward pass: given input X , weights W , and bias b , compute the output.
67. Write a program that implements softmax on a batch of 5 samples with 3 classes each and verifies each row sums to 1.
68. Write a program that manually one-hot encodes a label array of 20 samples with 4 classes.
69. Write a program that standardizes a dataset to zero mean and unit variance, then reverses the standardization to recover the original values.
70. Write a program that computes the pairwise distance matrix between all points in a dataset of 50 samples and implements a K-Nearest-Neighbors classifier using it.
71. Write a program that implements Linear Regression from scratch using the Normal Equation and tests it on a small synthetic dataset.
72. Write a program that implements gradient descent for Linear Regression from scratch, updating weights iteratively over 100 steps.
73. Write a program that implements Logistic Regression's forward pass and computes binary cross-entropy loss for a small dataset.
74. Write a program that implements PCA from scratch, reducing a dataset from 5 dimensions to 2 dimensions.
75. Write a program that generates weights for a neural network layer using Xavier initialization.
76. Write a program that implements a mini-batch generator: given a dataset of 1000 samples, yield random batches of 32 samples until the whole dataset has been used once.
77. Write a program that implements a Dropout layer: given an input array and a keep probability, randomly zero out values and scale the remaining ones.
78. Write a program that implements ReLU, Leaky ReLU, and their derivatives, and tests all three on the same input array.
79. Write a program that computes a confusion matrix from predicted and actual binary labels.
80. Write a program that computes accuracy, precision, recall, and F1-score from predicted and actual labels.
81. Write a program that detects and removes outlier rows from a dataset using the IQR method, applied independently to every column.
82. Write a program that implements Min-Max scaling and Z-score standardization as two separate reusable functions, and applies both to the same dataset.

83. Write a program that builds a simple 2-layer neural network's forward pass: input, hidden layer with ReLU, output layer with softmax.
84. Write a program that implements backpropagation for a single-layer neural network trained on a small binary classification dataset.
85. Write a program that implements K-Means clustering from scratch, including centroid initialization, point assignment, and centroid updates, repeated until convergence.
86. Write a program that generates a synthetic image as a 3D array and converts it from Height-Width-Channels format to Channels-Height-Width format.
87. Write a program that implements an im2col function that converts image patches into columns for a convolution operation.
88. Write a program that implements 2D convolution from scratch, applying a 3x3 filter to a grayscale image.
89. Write a program that implements max pooling and average pooling from scratch on a 2D feature map.
90. Write a program that computes a moving average over a time-series array without using any explicit Python loop.
91. Write a program that generates synthetic time-series data and creates sliding window sequences from it.
92. Write a program that implements gradient clipping by global norm on a set of gradient arrays.
93. Write a program that computes the Jacobian matrix of a simple vector-valued function.
94. Write a program that implements a basic Naive Bayes classifier's probability calculations for a small dataset with continuous features.
95. Write a program that implements cross-validation splitting, generating 5 folds where each fold is used once as validation data.
96. Write a program that implements data augmentation for a batch of images: random flip, 90-degree rotation, and brightness adjustment.
97. Write a program that implements weighted random sampling for an imbalanced dataset, so minority class samples are chosen more often.
98. Write a program that implements a recommendation system using cosine similarity between user vectors.
99. Write a program that implements SVD-based image compression, reducing the rank of an image matrix and reconstructing an approximate version.
100. Write a program that builds an end-to-end pipeline: load a synthetic dataset, handle missing values, scale features, split into train/test, train a from-scratch logistic regression model using gradient descent, and evaluate its accuracy.